Chenghao Li

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EDUCATION

Bachelor of **Computer Science**, Sept2019 – June2023

Oregon State University, Major GPA: 3.65 Corvallis, OR

- **Skill Set**: Unreal Engine, C++, Python, OpenGL/OpenCL, Vulkan, GLSL/HLSL shaders, CUDA (academic), OpenXR. GPU/Parallel, Unreal Insights, and platform performance tools.

PROFESSIONAL EXPERIENCE

Video Game Producer/Lead Engineer Feb 2025 – Current

Ylem Studio

Hustle (VR Game https://www.ylemstudio.com/)

- Owned the roadmap, scope, and milestones for a 4-person VR team, delivering a Quest 3 vertical slice for GDC 2026 on schedule, with stable 90 FPS and 4x MSAA using OpenXR and Unreal Engine.
- Built LLM copilot-friendly core game systems and collaborated with artists, engineers, and designers, translating the vision into clear tasks, priorities, and acceptance criteria. Applied Vulkan TonemapSubpass to support post-process.
- Led Agile delivery using Jira: managed the backlog, ran sprint planning/standups/reviews, tracked dependencies/risks, and cleared blockers to keep the team moving.

Unreal Engine Developer July 2023 - Jan 2025

Gamma Interactive Remote

Published title: X8 (VR Game https://www.meta.com/experiences/4518172508241377/)

- Collaborated with artists to implement gameplay functionality. Provide support for UE visual effects such as volumetric fog. integration and modification. Manage input systems related to throwing actions.
- Fixed bugs related to multiplayer games and optimized the performance based on Oculus Quest performance.

US Navy Aircraft Carrier Simulation project (VR/mobile)

- Build a VR application by unreal engine that allows multiple player collaborated together to finished Aircraft Carrier task, such as guide Helicopter drop the cargo on the deck and launching the F35 jet from a catapult.
- Communicated with the Navy agents who don't have a tech background, applied Agile development through fast iterative.
- Port to mobile platform, optimization for iOS and Android.

Macht(VR Game https://playmacht.com/)

- Build a custom movement component that supports VR character, server correction, and multiplayer compatibility
- Created customizable shader effects applied to characters that can interact with gameplay.

Research Assistant May 2022 - Aug 2023

NVIDIA Graphics & Image Technology Laboratory Corvallis

- Developed OpenXR runtime app using Unreal Engine to support Computer and Human interaction science research, focusing on VR education and training. Deployed the application to mobile(Android) and desktop(Windows) platforms, with latency optimization.
- Secured EDA (U.S. Department of Commerce) funding (award #07-79-07914) for an interactive VR forest simulation system, following a successful demonstration to the Oregon Department of Forestry.
- Implemented a VR education solution, using it in university classes and enhancing 200+ civil engineering students' grasp of building structures and force distribution, with OpenXR hand tracking.

SELECTED PROJECTS

Vulkan Ray-Tracing (https://github.com/lch9527/Vulkan) February 2023

- Built a Vulkan RTX ray tracer in C++/GLSL using VK_KHR_ray_tracing_pipeline,
- Implemented hardware Ray-Triangle Intersections by generating triangle sphere geometry, building a BLAS from mesh triangles, and constructing a TLAS with per-atom instances plus a reflective sphere.
- Developed the Firing Rays render path with camera ray generation, traceRayEXT / vkCmdTraceRaysKHR, dynamic TLAS updates, multi-light shading, reflections, and storage-image output copied to the swapchain.